Jonathan J. Hernandez, Ph.D.

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Qualifications Profile

- Biotechnologist and researcher in biopharma, stem cells, and acellular products for medical and cosmetic purpose, who provides fair-balanced, objective, and scientific information, and appropriate product education to investors, stakeholders, and clinicians.
- Experience in upstream bioprocess optimization including: scale up (bioreactors), process control, robustness and tech transfer for the culture, expansion, cryopreservation and banking of human mesenchymal stem cells derived.
- Unique skill set in innovating and building research and development laboratories from the ground up. Deep understanding of the drug/product development process, related compliance, and good manufacturing practices (GMP).
- Founded a startup company (Signature Biologics) including the full build of a research and development (R&D) laboratory. Hired and trained personnel and cultivated a successful product development function from the ground up.

Core Competencies

- Project Management
- Reporting & Documentation
- Feasibility Analysis
- Clinical Pharmacology & Toxicology
- Data Gathering & Analysis
- Current GMP (cGMP) Regulations
- Product Early Development
- Vendor Management
- Stakeholder Liaising
- Drug Discovery & Potency Testing
- Upstream bioprocess
- Translational Research
- Team Building & Leadership
- Training & Development
- Coaching & Mentoring

Scientific Expertise

- Adult Stem Cell Manufacturing
- Drug Metabolism/Pharmacokinetics
- Clinical Pharmacology & Toxicology
- Skin care product development and formulation
- Wound healing umbilical cord tissue-derive products.
- Tissue Bioengineering
- 2D Culture Approach
- Human-Induced Pluripotent Stem Cells Derivation
- Development of In Vitro Models
- Flow Cytometry
- Recombinant Protein/Proteomics
- Electrophysiology
- Secretome/Extracellular Vesicles

Career Highlights

- Created and validated 8 different in vitro models to test product potency and toxicity (Signature Biologics, LLC).
  - Developed in vitro models to test their effectiveness in the immunomodulation and restoration of musculoskeletal degenerative diseases such as Duchenne's muscular dystrophy (DMD).
- Generated up to $3M in sales within 15 months through marketing, sales, and R&D leadership (Signature Biologics, LLC).
- Contributed to section 8 of submitted Investigational New Drug (IND) package accepted by FDA in 2021 (Signature Biologics, LLC).
  - The IND outlines the use of an open-label, dose-ranging study to evaluate the safety, tolerability, preliminary efficacy, and dose effect of Signature Cord Prime in patients with symptomatic osteoarthritis in the knee.
- Accelerated new cellular product development after releasing peer-reviewed publication as VP of R&D (Signature Biologics, LLC).
- Completed new R&D lab build (5K square feet) and 3,900-square-foot ISO 8 cleanrooms; built storage and extra rooms for microbiology, cloning, and reagents preparation and sterilization (Signature Biologics, LLC).
- Established project management system for weekly/monthly data generation reports per project as well as financial/expense statements; saved capital expenditures and generated valuable data focused on simplicity and fast-paced experiments.

Professional Experience

Signature Biologics, LLC – Irving, TX 2018 - October 2022

Chief Research & Development (R&D) Officer

Promoted from Director to VP to Chief to undertake increased responsibility within the company. Direct personnel in experiments, scientific research, and clinical trials for the improvement of the human condition through placental-derived and stem cell therapy. Develop and characterize various human cells, tissues, and cellular/tissue-based products (HCT/Ps) regulated under 21 CFR 1271.3(d)(1) and Section 351 of the Public Health Service (PHS) Act; utilize HCT/Ps to target articular cartilage lesions, DMD, Pulmonary sarcoidosis and Frederic ataxia. Oversee hiring and development of a research technician.

Select Achievements

- Finalized and validated production process of secretome product obtained from mesenchymal stem cells isolated from a human umbilical cord (MCSs); utilizing current good manufacturing practices (cGMP) and xeno-free supplies.
- Sourced and gathered all preclinical data for submission of Section 8 of an IND for treating DMD using Signature Biologic's MSCs.

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- Earned orphan drug designation from FDA for utilizing and testing Signature Biologic's MSCs secretome product in a clinical trial for pulmonary sarcoidosis.

VP of Research & Development (R&D) – Pre-Clinical & Tech Transfer, 2020 – 2022

Created and executed research strategies to advance products from discovery into clinical development. Served as bridge between ideation, product development, and clinical research. Directed all scientific, technological, and research operations focusing on the development of placental-derived tissue and cellular/acellular products. Developed and characterized biological products by performing pre-clinical testing and demonstrating mechanisms of product actions, toxicity, dose-response, and application. Led and mentored internal development capability, spearheaded projects, and cultivated a lean, effective, and efficient infrastructure.

Select Achievements

- Initiated clinical trial phase 1 of Signature Biologic's first-ever FDA-classified 351 HCT/P product, Signature Cord Prime.
- Hired, trained, and developed 4 personnel including a research technician and senior scientist.
- Increased value of startup company by 25% via product development, characterization, and implementation of in vitro potency models for release criteria; proved mechanism of action of Signature Cord Prime.
- Spearheaded development of second-generation cellular (Product 2) and acellular (Product 3) FDA-classified 351 HCT/P .
- Submitted data for an orphan drug designation to the FDA for treatment of pulmonary sarcoidosis using mesenchymal stem cell-derived secretome.
- *Developed, formulate, and validate the new line of skin care products for micro needling procedures for scaring (UC-Derm and Amnio-Derm)*

Director of Research & Development (R&D), 2018 – 2020

Created the cGMP manufacturing process for various HCT/Ps. Oversaw clinical research organizations (CROs) and contract manufacturing organizations (CMOs) to execute outsourced product manufacturing. Communicated with vendors using strong understanding of regulatory requirements for pharmaceutical development. Set strategy for the discovery and pre-clinical research efforts, partnering across major discovery groups to address unmet medical needs and evangelize commercial potential of therapies.

Select Achievements

- Developed and transferred tech to cGMP lab for Signature Biologic's first-ever, FDA-requested-for-drug-designation-under-21 CFR 1271.10(a) 361 HCT/P product, Signature Cord.
- Facilitated field sales and marketing events for visitors and conference attendees by providing scientific materials that supported characteristics and mechanisms of product value within the competitive market.
- Recruited, hired, trained, and developed the company's first 2 research assistants (entry-level and mid-level).
- Successfully completed design and build of 800-square-foot R&D laboratory for molecular biology, biochemistry, and proteomics.
- Built a 300-square-foot ISO 8 cleanroom equipped for cell culture and fluorescent microscopy.

Additional Experience

University of Michigan, Department of Internal Medicine – Ann Arbor, MI 2018

Postdoctoral Research Associate – Division of Cardiovascular Medicine

- Executed research studies in pathophysiologic and mechanistic basis of familial hypertrophic cardiomyopathy, caused by mutations in myosin binding protein c (MyBP-C), using micro printing as a culture platform for human induced pluripotent stem cells derived cardiomyocytes (hiPS-CM) technology.

University of Wisconsin – Madison, Department of Pediatrics – Madison, WI 2016 – 2018

Postdoctoral Research Associate – Department of Cardiovascular Medicine

- Studied molecular and contractile mechanisms of familial hypertrophic cardiomyopathy (MyBP-C mutations) and DMD utilizing engineered cardiac 3D tissue constructs (ECTs) technology.
- Made ECTs; edited MyBP-C gene using CRISPR/Cas9 to obtain allogenic controls and induce mutations in WT hiPSCs.
- Performed simultaneous recordings of force and calcium handling on ECTs made with healthy, mutant, and CRISPR/Cas9 mutation induced hiPS-CM

University of Wisconsin – Madison – Madison, WI 2010 – 2016

Research Assistant

- Analyzed the molecular and cellular phenotype the mutation H2464D in the cardiac Ryanodine Receptor (RyR2) that causes mishandling of calcium in cardiac cells leading to familial catecholaminergic polymorphic ventricular tachycardia (CPVT). Recombinant RyR2, patient-specific hiPS-CM and a transgenic mouse were the in-vitro and in-vivo models developed, validated, and utilized for this project.

Education

Ph.D., Human Physiology & Regenerative Biology – University of Wisconsin-Madison, School of Medicine
Master of Science, Molecular Physiology – University of Wisconsin-Madison, School of Medicine
Master of Science, Higher Education – University of Panama, School of Education
Bachelor of Science, Biotechnology – University of Panama-San Martin, School of Biotechnology